

Question	Marking Guidance	Mark	Comments
04.1	1. Low respiration; 2. More growth/biomass/colonisation;	2	1. Accept less energy lost in respiration 2. Allow examples of more carbon-containing molecules eg glucose

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04.2	1. Less nitrification OR Fewer/less active nitrifying bacteria; OR Nitrification/nitrifying bacteria require oxygen/aerobic conditions; 2. (Less) oxidation/conversion of ammonium (ions) to nitrite (ions) and to nitrate (ions); 3. More denitrification OR More/more active denitrifying bacteria OR Denitrification/denitrifying bacteria do not require oxygen OR Denitrification/denitrifying bacteria require anaerobic conditions; 4. (So more) nitrate (ions) reduced/converted to nitrogen (gas);	2 max	2. Order must be nitrite then nitrate 2. Accept ammonia for ammonium ions 2. Accept correct chemical formulae for ions, eg there will be little oxidation/conversion of $\text{NH}_4^+ \rightarrow \text{NO}_2^- \rightarrow \text{NO}_3^-$ 2. Ignore 'breakdown' for oxidation/conversion 4. Accept correct chemical formulae eg So more NO_3^- reduced/converted to N_2 ;

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04.3	1. Assumed that height is (directly) proportional to biomass; 2. (Plants may put biomass into) other named aspect of growth (other than height) OR Height does not include the roots OR Some increase in height results from water gain;	2	1. Accept descriptions of 'is proportional to', eg correlates to, is equivalent to 2. Examples of other named aspects of growth could include root growth, flower/seed/fruit formation, lateral growth, wider leaves

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04.4	1. Answer of 12 days = 2 marks;; 2. $12.16 (12.15774433) = \mathbf{1 \text{ mark}}$ OR 4 days (used 387 and 268, ie not calculated starting length) = 1 mark;	2	
